

**Problem # 1:** Multiple choice question. [25 x 1pt]

- 1) If $f_1(n) \in O(g_1(n))$ and $f_2(n) \in O(g_2(n))$, then $f_1(n) + f_2(n) \in$
 A. $O(g_1(n))$ B. $O(g_2(n))$
 C. $O(\max\{g_1(n), g_2(n)\})$ D. None of the above

- 2) The best-case time complexity of sequential search is
 A. n B. 1 C. n^2 D. $\log n$

- 3) Given the below sorted array:

$\frac{6}{2} = 3$
 $\frac{18}{2} = 9$
 $\frac{16}{2} = 8$
 $\frac{16}{2} = 8$
 $\text{arr}[] = \{1, 14, 16, 20, 24, 26, 30, 40, 45, 49, 50, 55, 70, 97, 102, 107, 111\}$

To determine if $x = 32$ exists in the above array using the binary search algorithm, the numbers compared in order with x are:

- A. 45, 24, 26, 30, 40 B. 45, 20, 26, 30, 40
 C. 45, 20, 24, 26, 40 D. None of the above

- 4) Linked lists allow random memory access.
 A. True B. False

- 5) The every-case time complexity of Prime's algorithm is
 A. $\Theta(n^2 * \lg n)$ B. $\Theta(\lg n)$ C. $\Theta(n^2)$ D. None of the above

- 6) The solution of the recurrence in the table below is:

$$c\left(\frac{n-1}{2}\right) + d$$

$$8\left(\frac{n-1}{2}\right) + 5$$

$$T(n) = T(n-1) + 8n, T(1) = 5$$

$$4n^2 - 4n + 5$$

- A. $8 * n - 2$ B. $\log n$
 C. $4 * n^2 + 4 * n - 3$ D. None of the above

- 7) If the size of an array is 1024, then the total number of comparisons done by binary search and sequential search are respectively.

- A. 10, 1024 B. 11, 1024 C. 1024, 10 D. 1024, 1024

- 8) The big theta of μ
 $a = b^{\mu}$
 $8 = 2^3$
 $8 = 2^3$

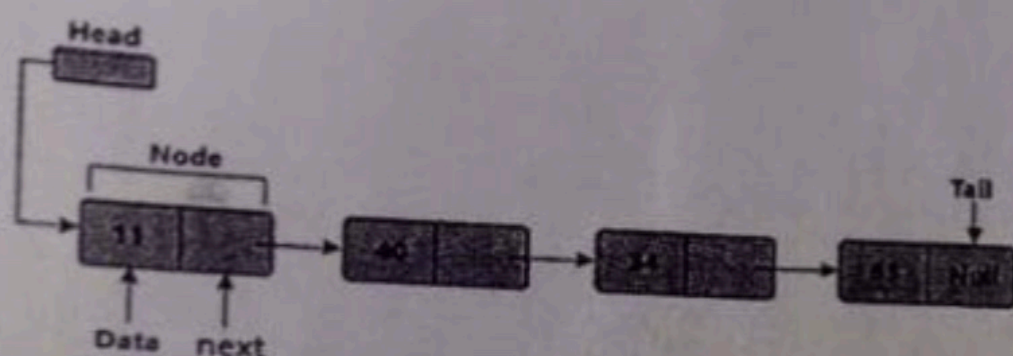
$$F(n) = 8F\left(\frac{n}{2}\right) + n^3$$

- A. $n^2 * \log n$ B. n^2 C. $n^3 * \log n$ D. None



9) Use the following LinkList to determine the value of following instructions

```
node <T> *q;
q = head->next->next;
cout<<q->next->data;
```



A. 65

B. 40

C. 34

D. 11

10) If the MAX_SIZE is the size of the array used in the implementation of the circular queue. How is the rear manipulated while inserting an element in the queue?

A. $\text{rear} = (\text{rear} \% 1) + \text{MAX_SIZE}$

B. $\text{rear} = \text{rear} \% (\text{MAX_SIZE} + 1)$

C. $\text{rear} = (\text{rear} + 1) \% \text{MAX_SIZE}$

D. $\text{rear} = \text{rear} + (1 \% \text{MAX_SIZE})$

11) In the linked list implementation of a queue, where does a new element be inserted?

A. At the head of the link list.

B. At the tail of the link list.

C. At the center position in the link list.

D. None of the above.

12) Which of the following is true about the linked list implementation of the stack?

A. In a push operation, if new nodes are inserted at the beginning of the linked list, then in a pop operation, nodes must be removed from the end.

B. In a push operation, if new nodes are inserted at the end, then in a pop operation, nodes must be removed from the beginning.

C. (A+B)

D. None of the above

13) Class and function templates allow the programmer to write one code for all data types and help in generic programming.

A) True

B) False

14) A circular queue is implemented using an array of size 10. The array index starts with 0, the front is 6, and the rear is 9. The insertion of the next element takes place at the array index

A. 0

B. 7

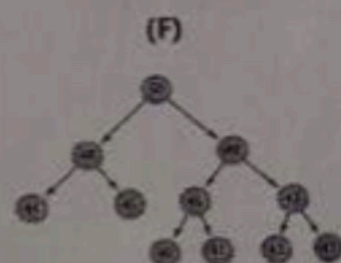
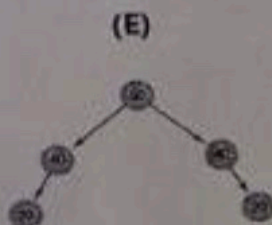
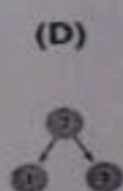
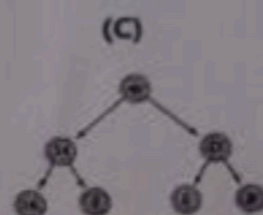
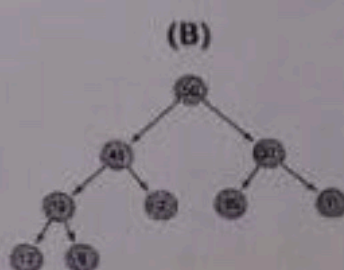
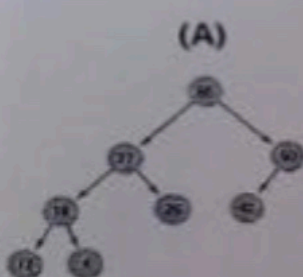
C. 9

D. 10

Size = 10
front = 6
rear = 9



20) Which of the following is a Complete Binary Tree?



- A. (B, C, and F) ✗
C. (A, B, C, and F) ✗

- ✗ B. (D, E, and F)
D. (B, C, and D) ✗

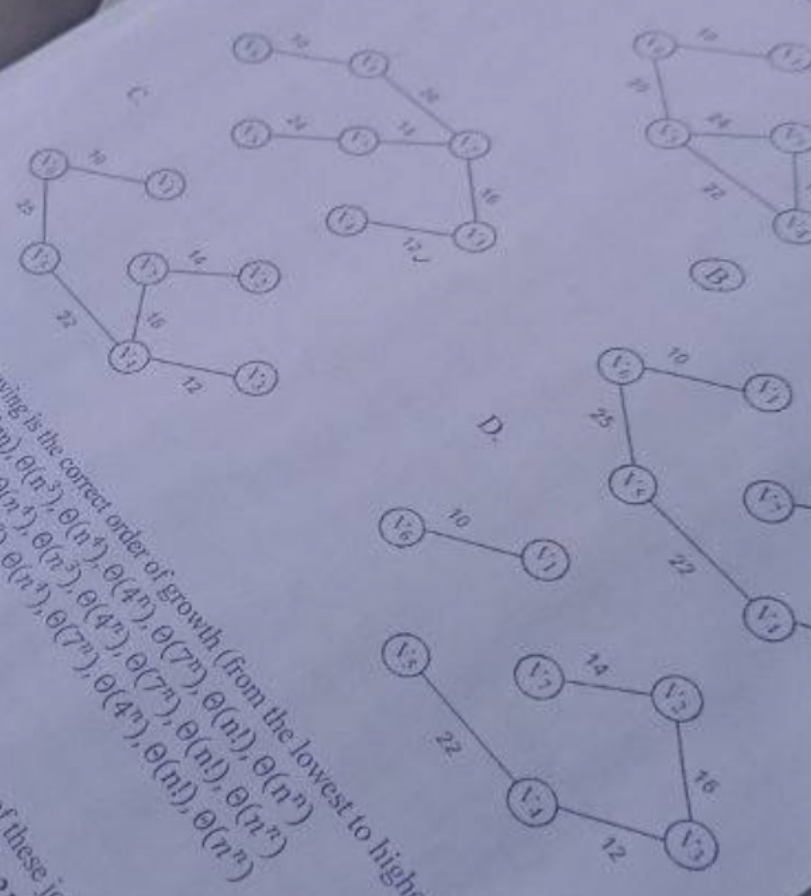
21) What does the following function do?

```
Fun (T num)
{ node <T>*q;
  q = new node<T>;
  q->data = num;
  q->link = head;
  head = q;}
```

- A. Insert at the end of link list
B. Insert at the middle of link list

- C. Insert at the begging of link list
D. insert to the empty link list

4. Which of the following is the correct order of growth (from the lowest to highest)?



- 19) Which of the following is the correct order of growth (from the lowest to highest)?
- A. $\theta(n^2)$, $\theta(n)$, $\theta(n \log n)$, $\theta(n^3)$, $\theta(n^4)$, $\theta(7^n)$, $\theta(n!)$, $\theta(n^n)$
 - B. $\theta(n)$, $\theta(n^2)$, $\theta(n \log n)$, $\theta(n^3)$, $\theta(n^4)$, $\theta(7^n)$, $\theta(n!)$, $\theta(n^n)$
 - C. $\theta(n)$, $\theta(n^2)$, $\theta(n \log n)$, $\theta(n^3)$, $\theta(n^4)$, $\theta(7^n)$, $\theta(n!)$, $\theta(n^n)$
 - D. None of the above

*** Suppose there are five jobs and the service time of these jobs are $t_1 = 20$, $t_2 = 14$, $t_3 = 9$, $t_4 = 22$, and $t_5 = 7$, answer the questions 20 and 21:

- 20) The job's schedule that minimizes the total time spent in the system is:
- A. [1, 2, 3, 4, 5]
 - B. [5, 3, 2, 1, 4]
 - C. [5, 3, 4, 1, 2]
 - D. None of the above
- 21) The total time spent in the system for the optimal schedule is:
- A. 165
 - B. 185
 - C. 175
 - D. None of the above



Problem # 2: Given the two sequences "ACDBAC" and "ACDAACD", complete the table below to find the Longest Common Sequence (LCS) using Dynamic Programming (DP). [6pts]

j	0	1	2	3	4	5	6
i		A	C	D	B	A	C
0	0	0	0	0	0	0	0
1 A	0	1	1	1	1	1	1
2 C	0	1	2	2	2	2	2
3 D	0	1	2	3	3	3	3
4 A	0	1	2	3	3	4	4
5 A	0	1	2	3	3	4	4
6 C	0	1	2	3	3	4	5
7 D	0	1	2	3	3	4	5

LCS is ..AC.D.AC.....

Problem # 3: Answer the following questions supposing a data file has characters and frequencies shown in the table below. If Huffman coding is used: [10 pts]

Character	Frequency
A	33 ✓
B	24 ✓
C	10 ✓
D	12 ✓
E	42 ✓
F	13 ✓
G	26 ✓